AIM

To assess the prescribing patterns, dispensing frequency, and consumption of antihypertensive medications in order to inform drug forecasting, enhance monitoring practices, and optimize resource allocation.

SPECIFIC OBJECTIVES

To describe the demographic characteristics(age, gender) of patients receiving antihypertensive medications during the study period (Feb.2024 – September 2024)

To evaluate the types and strengths of antihypertensive drugs prescribed and dispensed during the study period, and to identify the most frequently prescribed antihypertensive drugs.

To identify cases of early refills for each antihypertensive medications and determine the amount lost due to early refills

To determine the consumption rate of the antihypertensive medications in the clinic

To provide a solution based on findings

PROCEDURE

The dispensing data of antihypertensives from February 2024 to September, 2024 was abstracted from the record book and entered into a spreadsheet.

Information entered- Personal number, Gender, Age, Date of dispensing, Antihypertensives dispensed and the quantity(0.5 for 2 weeks, 1 for one month, 2 for two months and so on)

The data was cleaned and analyzed using Microsoft Excel and Python packages for data and statistical analysis (Numpy, Pandas, Scipy, Matplotlib)

FINDINGS

Demographic characteristics (gender) of patients receiving antihypertensive medications during the study period (February 2024 – September 2024).

Gender	Count
Male	2196
Female	1802

Summary

The clinic saw more male patients with hypertension than female patients during the study period

FINDINGS

Demographic characteristics (age) of patients receiving antihypertensive medications during the study period (February 2024 – September 2024).

Statistics	
Mean	63.75
Standard deviation	10.29
Median	65.00
Max	92.00
Min	15.00

Summary

The mean age of antihypertensive patients visiting the clinic during the study period is **64**.

FINDINGS

A comparative analysis of patients' gender and age- Independent sample t-test

Gender	Mean	T-statistics	P-value	Significance
Male	66.04	15.9901	0.0000	**Yes**
Female	60.97			

Summary

The male patients had a significantly higher mean age than the female patients, suggesting that generally the clinic sees more older male patients with hypertension than female patients.

FINDINGS

Evaluation of the types of antihypertensive drugs prescribed and dispensed

Medication	Two weeks	One month	Two months	Three months	Four months	Total
Amlodipine 10mg	130	2233	6	0	0	2369
Losartan 50mg	44	737	9	0	0	790
Lisinopril 10mg	61	575	5	0	1	642
Amlodipine 5mg	44	476	0	0	0	520
Indapamide 1.5mg	16	465	5	0	0	486
Bendroflumethiazide 2.5mg	2	462	6	2	1	474
Bisoprolol 2.5mg	28	335	3	2	4	372
Lisinopril 5mg	0	282	0	0	0	282
Normoretic	11	219	6	0	0	236
Losartan 100mg	26	207	0	0	0	233
Telmisartan 80mg	5	121	3	0	0	129
Spirinolactone	11	100	4	1	0	116
Hydrochlorothiazide 25mg	40	31	0	0	0	71
Valsartan 160mg	0	67	1	0	0	68
Nifedipine XL 30mg	0	58	4	1	0	63
Nifedipine 20mg	1	52	2	2	1	58
Methyldopa 250mg	0	39	7	2	7	55
Candesartan 16mg	0	53	1	0	0	54
Aprovasc	0	49	2	1	0	52
Co-diovan	1	47	3	0	1	52

Summary

Amlodipine 10mg were prescribed and dispensed **2369** times within the timeframe this study was conducted

FINDINGS

On the average what percentage of hypertensive patients who visit the clinic per day collect a particular drug, say amlodipine ?

What is the probability that an anti-hypertensive patient in the clinic takes a particular drug, say amlodipine?

Medication	Average Daily Percentage (%)	Probability
Amlodipine 10mg	59.4	0.6
Lisinopril 10mg	20.2	0.2
Losartan 50mg	16	0.2
Indapamide 1.5mg	12.5	0.1
Bendroflumethiazide 2.5mg	12.4	0.1
Amlodipine 5mg	12	0.1
Bisoprolol 2.5mg	9.7	0.1
Losartan 100mg	7.2	0.1
Lisinopril 5mg	6.4	0.1
Normoretic	5.7	0.1

- *Out of every 100 hypertensive patients that visit the clinic in a day, about 60 of them collect Amlodipine 10mg*
- *The probability that a hypertensive patient that walks into the clinic in a day collects Amlodipine 10mg is 0.6*

FINDINGS

Day Of Week	Count
Monday	851
Tuesday	825
Thursday	823
Wednesday	759
Friday	740

Summary

Antihypertensive medications were most filled on Mondays and least on Fridays

FINDINGS

Detection of cases of early refills

Medication	Early Refills
Amlodipine 10mg_Early	456
Losartan 50mg_Early	119
Lisinopril 10mg_Early	102
Indapamide 1.5mg_Early	82
Bendroflumethiazide 2.5mg_Early	68
Bisoprolol 2.5mg_Early	67
Amlodipine 5mg_Early	51
Losartan 100mg_Early	39
Normoretic_Early	32
Spirinolactone_Early	22
Lisinopril 5mg_Early	22
Telmisartan 80mg_Early	20
Nifedipine XL 30mg_Early	15
Candesartan 16mg_Early	11
Nifedipine 20mg_Early	10
Aprovasc_Early	7
Labetalol hydrochloride_Early	7
Atenolol 100mg_Early	6
Coveram plus(10mg/2.5mg/10mg)_Early	6
Valsartan 80mg_Early	6

Summary

There were 456 cases of early refill of Amlodipine 10mg

FINDINGS

Tablet leftovers (The amount of medicine(tablets) remaining with the patients as of the time he/she came for the early refill) and total packs lost due to early refill

Medication	Quantity in a pack	Total tablet lost in early refill	Total packs lost
Amlodipine 10mg	30	4708	157
Losartan 50mg	30	1914	64
Lisinopril 10mg	28	1168	42
Indapamide 1.5mg	30	910	30
Bisoprolol 2.5mg	28	984	35
Bendroflumethiazide 2.5mg	28	810	29
Amlodipine 5mg	30	603	20
Normoretic	100	356	4
Lisinopril 5mg	28	204	7
Spirinolactone	100	184	2
Telmisartan 80mg	30	160	5
Nifedipine XL 30mg	30	148	5
Aprovasc	28	135	5
Nifedipine 20mg	100	124	1
Labetalol hydrochloride	30	122	4
Candesartan 16mg	28	109	4
Coveram plus(10mg/2.5mg/10mg)	30	85	3
Co-diovan	28	70	3
Atenolol 100mg	28	65	2
Valsartan 80mg	28	63	2

Summary

For the period of this study, 4708 tablets which is about 157 packs of Amlodipine by 30 has been lost to early refill

FINDINGS

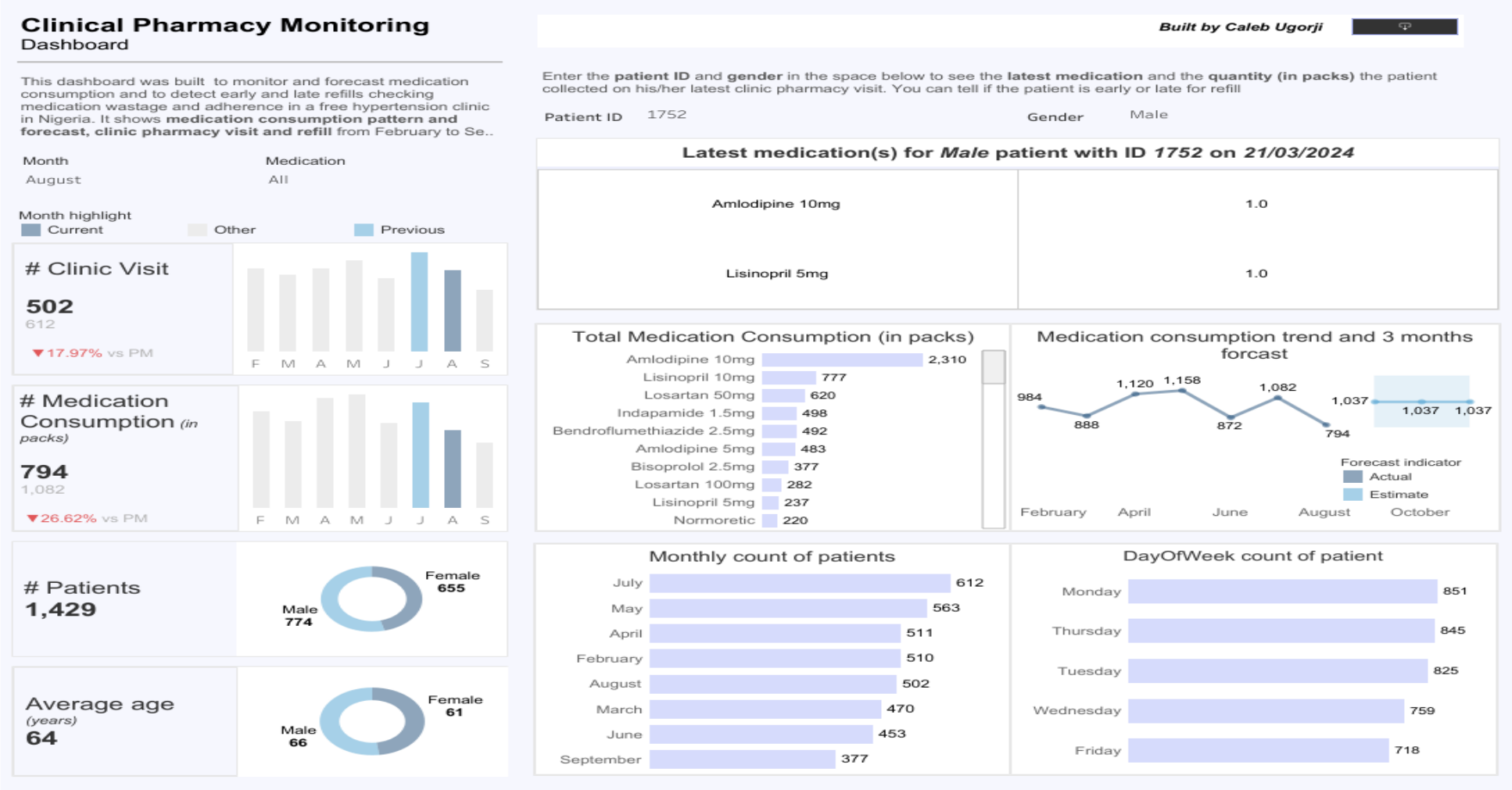
Identification of antihypertensive medication with the highest consumption- *Total packs of the medication consumed within the time frame of the study*

Medication	Quantity in a pack	Total packs	Average Monthly consumption(packs)
Amlodipine 10mg	30	2159	270
Losartan 50mg	30	1,519	190
Lisinopril 10mg	28	1184	148
Indapamide 1.5mg	28	712	89
Amlodipine 5mg	30	498	62
Bendroflumethiazide 2.5mg	28	485	61
Bisoprolol 2.5mg	28	479	60
Lisinopril 5mg	28	130	16
Hydrochlorothiazide 25mg	100	78	10
Normoretic	100	69	9
Telmisartan 80mg	30	69	9
Spirinolactone	100	67	8
Valsartan 160mg	28	58	7
Nifedipine XL 30mg	30	56	7
Methyldopa 250mg	100	56	7
Nifedipine 20mg	100	55	7
Candesartan 16mg	28	43	5
Aprovasc	28	39	5
Co-diovan	28	37	5
Coveram plus(10mg/2.5mg/10mg)	30	35	4

Summary

2159 packs of amlodipine by 30 were consumed within the time frame of this study. The average monthly consumption of Amlodipine 10mg is **270 packs**

SOLUTION

1. A system that identifies in real time the lastest patient visit, the medication collected and the quantity given

SOLUTION- CHECK THE SOLUTION FOLDER